Functional Dependencies and Normalisation

txt2tex example

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$[\mathit{EmpID}, \mathit{EmpName}, \mathit{DeptName}, \mathit{HeadName}, \mathit{ProjID}, \mathit{ProjName}]$

The Universal Relation

We start with a single unnormalised relation that bundles every attribute together. The composite primary key is (empID, projID) – an employee can be assigned to many projects, and a project involves many employees.

$$
\begin{array}{|l}
\hline \textit{Universal} \\
\underline{\mathit{empID}} : \mathit{EmpID} \\
\mathit{empName} : \mathit{EmpName} \\
\mathit{dept} : \mathit{DeptName} \\
\mathit{deptHead} : \mathit{HeadName} \\
\underline{\mathit{projID}} : \mathit{ProjID} \\
\mathit{projName} : \mathit{ProjName} \\
\hline
\end{array}
$$

The relation is in 1NF: every attribute is single-valued and every row is uniquely identified by (empID, projID).

Functional Dependencies

We identify the following functional dependencies (FDs).

Full-key dependency (no issue):

empID, $\mathit{projID} \longrightarrow \mathit{empName}$, dept, deptHead, projName

Partial dependencies – empName, dept, and deptHead depend on empID alone; projName depends on projID alone. These violate 2NF.

$\mathit{empID} \longrightarrow \mathit{empName}$, dept, deptHead $\mathit{projID} \longrightarrow \mathit{projName}$

Transitive dependency – deptHead is determined by dept, not directly by empID. This violates 3NF even after removing the partial dependencies.

$\mathit{dept} \longrightarrow \mathit{deptHead}$

2NF: Remove Partial Dependencies

A relation is in 2NF when every non-key attribute depends on the whole primary key, not just part of it. We remove empName, dept, deptHead (which depend only on empID) and projName (which depends only on projID) into their own relations.

Remaining composite relation after 2NF lift – empID and projID are the composite primary key; all remaining attributes depend on the full key:

EmpProj2NF(empID, projID)

Employee attributes extracted into a separate relation:

EmpDept2NF(empID, empName, dept, deptHead)

Project attributes extracted into a separate relation:

Proj2NF(projID, projName)

EmpDept2NF is still not in 3NF: $dept \longrightarrow deptHead$ is a transitive dependency. deptHead is a fact about the department, not the employee.

3NF: Remove Transitive Dependencies

A relation is in 3NF when every non-key attribute depends directly on the primary key and on nothing else. We remove deptHead (determined by dept) into a Department relation and drop it from EmployeeProj.

Final 3NF decomposition – four relations:

Employee
$\underline{empID} : EmpID$
$empName : EmpName$
$dept : DeptName$

Department
$\underline{dept} : DeptName$
$deptHead : HeadName$

Project
$\underline{projID} : ProjID$
$projName : ProjName$

Assignment
$\underline{empID} : EmpID$
$\underline{projID} : ProjID$

Every non-key attribute now depends only on its own primary key. No partial or transitive dependencies remain. The decomposition is lossless: rejoining Employee, Department, Assignment, and Project on their shared keys reconstructs the original Universal relation exactly.